Do Now · retrieval

Procedures, functions and parameters

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

A local x=4 shadows a field x=9. Which is printed inside the method?

Q2

What is a base case?

Q3

What is a[2] for int[] a = {3,5,8,13}?

Q4

Starting x=1, apply x=x*2 three times. List the results.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

CALL — The caller supplies argument 6 to Double.

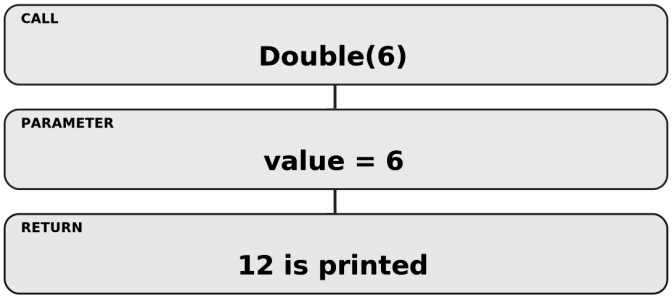
Core idea 2

PARAMETER — Parameter value receives 6, then value * 2 evaluates to 12.

Core idea 3

RETURN — The function returns 12. Console.WriteLine displays the result.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

procedure — A subroutine that performs actions and does not return a result.

function — A subroutine that returns a value.

parameter — A named input in the subroutine declaration.

argument — The actual value provided by the caller.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

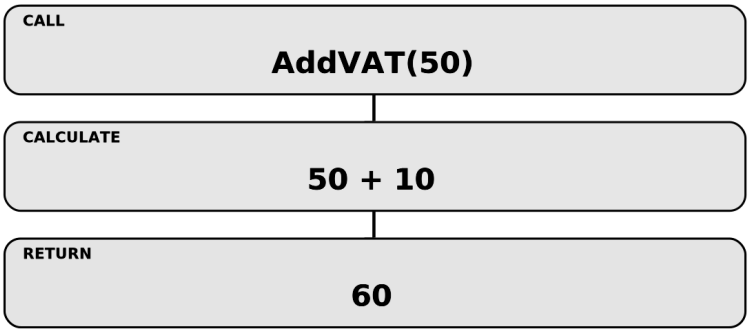
Problem

Trace AddVAT(50), which returns price + price / 5.

Follow the worked method

1. Predict first — The argument 50 becomes local parameter price = 50.
2. Model the result — Integer division 50 / 5 = 10; the returned value is 60.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
The argument 4 is assigned to n.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

A function Square(n) returns n*n. Find Square(5).

Question 2

Write a C# function IsEven(int n) that returns true when n is even.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why static int Show(int x) { Console.WriteLine(x); } is invalid. Correct it.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A C# method with return type void must return an integer.

Use key vocabulary from memory

1. Explain procedure in your own words.

2. Explain function in your own words.

3. Explain parameter in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.